

Lamina

Source-aware production methods for document workflows

Lamina Project

21 September 2026 — technical design and implementation status

Abstract

Producing a guide, brief or assessment from changing sources requires decisions about authority, context, division of work and repair. Lamina is open-source infrastructure for those decisions. Its current paths include a fixed guide builder, a separate method runtime and a source-aware production engine connected to a local app, CLI, exports and recovery. Source roles, assessment checks and optional PCM rendering cover specific output boundaries. This paper defines source ownership, context selection, dependency and resource edges, cache identity and repair in a task-shaped architecture. Synthetic scheduling and deterministic fixtures establish mechanics. The product hypothesis is that a trace from source decisions to delivered artifacts reduces correction work at matched quality; that outcome still needs a new-source study.

1 The production problem

An operating brief may draw on a current procedure, a superseded note, exceptions in a table and a document used only for style. A single model call can merge old and current rules or lose a condition. Sending every source to every writer costs more and still leaves attention uncertain; long-context retrieval varies by position and task [7]. Parsing and planning can also discard a condition before writing starts.

After a source changes, the owner needs to know which decisions depend on it, what can be reused and whether the corrected file reached its destination. Rebuilding everything costs time and risks new errors; editing final prose alone can leave a false upstream interpretation in place. Lamina treats a *production method* as the versioned transformation from a defined source set to an artifact, with decisions and repair paths retained.

Parallel LLM function calls have been studied [5]; graph workflows, durable state and interruptions already exist [2, 6]. The question here is whether source decisions and repair traces lower total correction work at matched output quality.

2 Evidence from several output forms

A differential reference needs one owner per retrieval target, useful variants and a findable answer. A spoken lesson needs lists, explanation, purposeful repetition and playable audio. An interactive case needs candidate-only findings, examiner answers and new information after actions or elapsed time. A written case adds a setter, question-specific keys and a blind solver limited to the candidate’s staged

view. A technical brief needs an argument that accounts for conflicting sources. They share source identity and revision machinery, but their representations differ.

Reading a source item and including it in an artifact are separate events. A quotation records provenance without establishing that the cited claim follows. File structure also differs from use: a valid script may produce poor audio, a case may be unrealistic, and a PDF may hide an illegible diagram. Each boundary needs its own check.

3 A task contract before a graph

The task contract is

$$\mathcal{T} = (G, A, S, P, O, Q, R), \quad (1)$$

where G is the goal, A the audience or operator, S the accepted sources, P their use policy, O the artifact and delivery surface, Q acceptance criteria and R limits on time, model, money, context and machines. Missing fields weaken what a completed run can claim; the tuple does not score quality.

A current specification may govern an operational rule; an older note may document conflict without current authority. A form exemplar may shape presentation while contributing no factual citation, and held-out questions stay outside generation. Lamina accepts authority, supplement, historical and form-exemplar policies for selected teaching sources. Factual sources enter reading; form exemplars enter bounded form context; held-out assessment sources stay excluded. Source identity, hash, role and locator accompany the decision. Policy changes alter the plan’s meaning even when source bytes do not change. Review still has to judge the model’s use of authority.

A prior question may be a form example while another is withheld for evaluation. A source manifest should distinguish discovery, inventory, reading, factual use, form use and withholding. Counts at one stage cannot stand for coverage at another.

Representation follows the job: guides need claims, conditions and causal steps; differential references need presentation-to-answer objects and variants; interactive cases need states, requests, findings, actions, consequences and marking opportunities. Figures and tables need modality-aware extraction. Review must then examine the appropriate unit: support for a document claim; answer family, exclusion and qualifier for a retrieval target; or reachable findings and treatment-linked observations in a case. A generic pass score cannot express these checks. The method should say which failure blocks delivery, flags review or opens a source-policy decision. A model reviewer has no independent authority over high-stakes factual claims.

Models may propose objects and relations such as same target, variant, contrast, prerequisite and conflict. Code owns stable IDs, locators, membership and limits. Ambiguous merges remain open. An object may be written, deferred or deliberately omitted according to the task. Every required field needs a consumer in writing, review or repair. Structural accounting should expose the original region and claimed transformation: one owner per extracted item can coexist with a merge that loses an exception.

4 Source windows and ownership

A reading job j receives a focal core C_j and neighboring halo H_j . It may use the halo to interpret the core, while owning extraction only from C_j . For readable accepted regions $S_1, \dots, S_n$, require

$$\bigcup_j C_j = S_{\text{readable}}, \quad C_j \cap C_k = \emptyset \text{ for } j \neq k. \quad (2)$$

This partition prevents duplicate ownership at boundaries. A region inventory must still identify unreadable figures, broken table order and missed clauses. Small windows reduce call size but can lose distant qualifiers; large ones increase repeated context and competition for attention. Choose core size, halo and bridge review for the source and task. Preserve IDs and short supporting spans, and let reviewers reopen the full source when a local packet is insufficient.

5 Methods and execution semantics

A method names applicability, exclusions, artifacts, worker roles, passed data, dependencies, resource limits and review policy. A task with unsupported source modality or acceptance criteria should reject the method. Current method files declare bounded JSON graphs run through a configured adapter; selection and revision remain manual.

For jobs V , four relations matter:

- A *hard-result edge* $u \rightarrow v$ makes v wait for and receive u 's result.
- A *shared-plan relation* gives jobs the same versioned editorial map without requiring finished prose from another writer.
- An *evidence-validity relation* marks work to reconsider after a source or decision changes; it does not require regeneration of every descendant.
- A *resource relation* assigns finite-capacity model, parser, renderer or review lanes without adding semantic information.

The generic runtime implements hard-result edges and resource lanes. Production uses local parallel reading, a shared route, parallel writing and review. Some jobs need earlier evidence; others need a finished prior section. Both patterns remain useful.

A short coherent file may suit source order. A shared topic board coordinates overlapping files but makes reconciliation a global cost and possible merge error. Chaining finished sections improves continuity while lengthening the critical path and later contexts. An editor over parallel drafts adds review cost. Choose among them using source topology, output form, conflicts, context budget and deadline. Keep a rejected alternative when it helps explain a later repair.

6 Context assembly as an explicit decision

Let I_v be candidate context items for job v , with protected subset P_v required by the task or a consequential claim. Item i uses b_i tokens in the adapter's tokenizer. With input budget B_v after

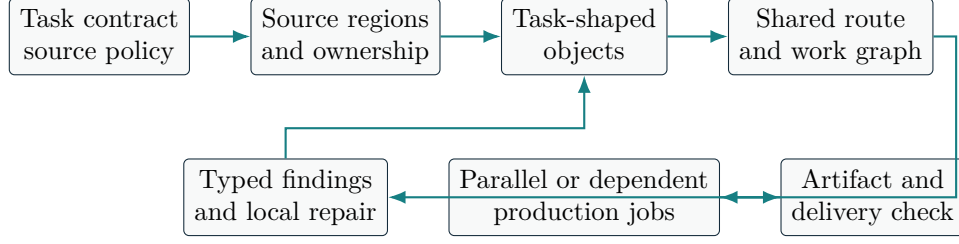


Figure 1: Intended source-aware production loop. The diagram separates semantic decisions from scheduling and delivery. Review may reopen a specific object or source decision; it need not restart every job. The fully connected loop is a design target, not a claim that all boxes are automated in the current release.

instructions and output reserve, choose X_v so that

$$P_v \subseteq X_v, \quad \sum_{i \in X_v} b_i \leq B_v. \quad (3)$$

Rank optional items for relevance. If P_v alone exceeds B_v , split the job or change scope. Record plan version, included source and object IDs, omitted counts and actual request size. A reviewer needs broader inventory access than a writer to find material lost during selection.

Context amplification is

$$A = \frac{\sum_{v \in V} \text{inputTokens}(v)}{\text{tokens}(\text{distinct accepted source text})} \quad (4)$$

Report A by stage with largest request, source coverage and accepted quality. Repeating a protected rule can prevent error; repeating whole sources can waste cost and attention. The method runner selects top-level task fields and direct dependency results, but does not yet perform the source-aware constrained choice in Eq. 3.

7 Concurrency and measured resource limits

For service times t_v , work $W = \sum_v t_v$ and longest hard-dependency path D , the ideal lower bound with p equivalent workers is

$$T \geq \max \left(D, \frac{W}{p} \right). \quad (5)$$

For lane r with capacity c_r and service work W_r , add its limit:

$$T \geq \max \left(D, \frac{W}{p}, \max_r \frac{W_r}{c_r} \right). \quad (6)$$

These classical work/span and capacity bounds assume compatible time units and a defined task graph [1]. They omit simultaneous resource demands, throttling, setup, queue waits, retries, human review and delivery. A global reconciliation call remains on D regardless of writer count. Waiting for earlier completed prose can reduce duplication while lengthening D ; a shared accelerator may serialize speech rendering and transcription. Batch arrivals and correlated calls can also defeat a steady-state queue model. Measure enqueue, start and end times, lane occupancy and latency tails.

Worker cap	Cold wall (s)	Peak handlers	Warm wall (s)	Warm calls
1	6.001	1	0.054	0
4	1.690	4	0.020	0
16	0.777	16	0.082	0
32	0.626	30	0.136	0

Table 1: Synthetic sleep workload on one local machine (Darwin arm64, Python 3.14.7, SQLite 3.53.4); one cold and one warm run per width. No model, human review, figure parsing or media delivery is in these times. Source: `benchmarks/results/production-widths.json`.

A checked-in fixed-graph benchmark ran 24 sleep-only extraction jobs, two global steps and six author/review pairs. At worker limits 1, 4 and 8, cold times were 2.412, 0.865 and 0.595 seconds; warm cache times were about 0.018–0.022 seconds with zero handler calls. The integrated production path needs its own measurement with models and accepted artifacts.

A second SQLite/cache run used 64 sleep-only readers, two serial global jobs and 16 author/review chains, totaling 98 handlers. Table 1 gives one local run per width. At width 32, peak overlap was 30 because of serial stages and chain dependencies. Warm calls stayed at zero, though bookkeeping made their wall times non-monotone.

8 Cache identity, review, and repair

A cache key must include the actual request: source and policy versions, instructions, selected task fields, dependency results, adapter identity and selected observations. Unrelated changes can retain work; a changed source meaning or review policy must invalidate affected results. A cache hit says the request matched, not that its answer is good. A reviewer verdict also becomes stale when the artifact or source it examined changes.

Findings should name the affected object, location, defect, source, severity, reviewer and artifact revision. Examples include an unreadable region, missing answer, false merge, unsupported claim, wrong condition, broken cross-section reference, candidate-answer leak or delivery failure. The finding can reopen parsing, representation, selection, writing, rendering or publication. Conflicting source authority may require a human decision.

The transitive dependency closure is a candidate recheck set, not an exact repair bill. For observation o , define

$$R(o) = (s_o, m_o, h_o, d_o), \tag{7}$$

where s_o is re-executed service seconds, m_o reported model or external charges, h_o human review minutes and d_o replacement deliveries. Keep the units separate unless a conversion is declared. Production supports named section revisions and one bounded repair and re-review; the fixed guide can block export on review failure. Routing findings to upstream causes and independently judging the final correction remain limited.

9 Experience and method reuse

A cache reuses unchanged job output. Method learning changes a future choice of source policy, representation, window, graph, context, reviewer or format. Lamina stores receipts and operator observations. An operator can attach a same-family observation ID to a later method node or production plan, where it enters context. Selection remains manual, and the effect on output has not been established.

Treat a method revision as a testable proposal: name the failure, changed decision, preserved behavior and added cost. Compare old and new versions on sources outside tuning, including a task where the favored approach should fail. Record losses. DSPy already studies metric-driven optimization of LLM pipelines [4]; Lamina’s question is whether task-specific source and repair decisions improve accepted documents under change.

10 Worked example: a disputed timeout

A fictional team is writing an export-worker brief. Its current specification sets a 90-second lease, renewal every 30 seconds while healthy, cessation of output after failed renewal, and reassignment only after expiry. A historical runbook expected 60 seconds and describes an incident where a second worker began while the first appeared to write. It does not establish that incident’s configured timeout or whether partial output was cleaned up. The team needs to check the contract before changing its retry timer.

Independent readers preserve both source claims with excerpts and conditions. Reconciliation applies the owner’s authority policy: the 60-second rule stays historical, and the incident’s cause stays unresolved. Behavior and operations writers use one evidence board in parallel. A reviewer sees both drafts and the board to find lost qualifiers or unsupported recommendations. The portable six-node technical-brief example validates this graph and context wiring; no model-written brief from it has been independently accepted.

If the current lease changes, its reader and the board must rerun. Both writers may need new context even when only one visibly states the number. The historical reader can remain cached when its source and task inputs match. A cleanup finding points to a source gap, so the brief should say to check the actual cleanup contract. A repair receipt should identify source version, reopened decisions, executed and reused jobs, review and delivered file. The portable example does not execute that entire correction journey.

A separate deterministic production fixture uses two original short sources and three sections. Its first run flags and repairs one section, with no remaining finding. Revising section three hits cache on six of eight requests; its write and review rerun, while the first two bodies remain byte-identical. Canned JSON responses test routing and reuse, not model judgment. Their millisecond timings are fixture timings.

11 Current implementation

The fixed guide builder ingests Markdown, text and extractable PDF text with hashes, units and locators. It reads with adjacent context, reconciles concepts, plans, writes, reviews and exports source-linked material. Validation checks ownership, quotations, membership and output shape.

Capability	Present mechanism	Status boundary
Source identity and policy	Hashes, ordered units, locators, authority/supplement/historical/form roles	Working for supported text; model’s use of authority still needs review
Fixed guide	Extract, reconcile, plan, author, review and export	Working application; one teaching route
Reusable method	User-declared JSON jobs, dependencies, lanes, cache and receipts	Working separately; semantic output rules external
Source-aware production	Bounded reading, shared route, section production and revision	Working local engine/API/CLI; semantic quality unproven
Retrieval targets	Opt-in guide/assessment catalog partitions extracted ideas into source-backed prompts and answer groups	Structural ownership checked; semantic equivalence unproven
Assessment check	Every assessment runs candidate-only solve, then separate judge with key and exact excerpts	Working local stage; no interactive administration or provider-memory guarantee
Audio delivery	Optional command adapter, PCM WAV check, hashes and local render lock	Working local file verification; no listened or ASR quality check
Experience	Operator-selected observations enter named future nodes	Explicit reuse; no automatic adaptation
Local delivery and recovery	HTML, Markdown, optional PDF, report, plan and project resume	Working locally; external publication separate
Typed upstream repair	Finding-to-cause routing across source, representation and output	Design target beyond section repair

Table 2: Status distinctions in the current checkout. Local end-to-end wiring and deterministic tests establish mechanics; they do not establish human-accepted model quality.

The separate method runtime runs user-declared JSON jobs with hard dependencies, selected task fields, capacity lanes, cached receipts and manually selected observations. It checks for JSON objects; declared shapes remain hints. Production reads factual teaching sources in owned windows, passes form exemplars separately, lets a model choose sections and representations, and writes and reviews under separate capacities. The optional guide/assessment retrieval stage gives each extracted idea one target and source-exact answer items. Assigned targets reach the guide or examiner artifact; omissions stay in the plan and report.

Production repairs a flagged section once and rechecks it. Named revisions reuse unchanged requests. The local project API persists state, marks unfinished projects interrupted after restart and resumes from plans and cache. CLI and Projects share the engine. Export produces Markdown, HTML, plan, report and optional PDF. Assessment keeps candidate and examiner files separate; every assessment runs candidate-only answers followed by key-and-evidence judging. A configured audio adapter ren-

ders ready podcast scripts to PCM WAV, checks file structure, duration and hashes, and records synthesis input. A local per-user file lock serializes uncached adapter calls.

12 Limits

Text extraction cannot recover arbitrary figures or prove that a citation entails a claim. Target ownership cannot validate semantic equivalence or material the reader missed. Separate candidate and judge calls do not erase a provider’s hidden state or administer a live case. PCM validation does not check spoken words, pronunciation or listener use. Production does not infer every cross-section dependency or route a missing idea through upstream parsing and planning. The method runtime remains a separate path without automatic method selection or demonstrated transfer. The current fixtures establish mechanics; judged output quality and comparative advantage require the study below.

13 Evaluation and competitive position

Compare time and total cost to a *quality-matched accepted artifact*, including operator work. Freeze a new permissioned corpus and task contract. Include a difficult table or figure, overlap, a contradiction and an omission risk. Independent reviewers inventory important source objects, conditions and acceptable omissions from rendered originals, keeping disagreements. Hold evaluation questions out of generation. Compare a careful source-order workflow and, where feasible, fully configured external products under the same task rules. Hold the model/provider fixed for an internal runtime comparison; across products, report the complete system result.

Record first useful draft and accepted accessible artifact times; source misses, false merges and unsupported claims; editorial usability; tokens, observed provider charges, human minutes, retries, failed attempts and delivery checks. Label estimated API-equivalent costs separately. Rotate paired run order and report median, tails, range and sample size. Then change a source, seed local and global findings, and ask a fresh operator to repair and deliver. Check whether the reduced repair set was safe. A fresh-agent run tests transfer without the original author.

Manus documents wide research and evolving project context [8, 9]. NotebookLM generates source-based audio [3]. LangGraph and Google’s ADK support graph workflows, including flexible and dynamic workflows in ADK’s current documentation [2, 6]. The possible Lamina advantage is narrower: an operator may be able to explain and repair a source-to-artifact decision with less lost work at matched quality. This remains unmeasured.

14 Conclusion

Lamina gives code responsibility for identity, source boundaries, scheduling, persistence and receipts while models read, plan and write. Its guide, method and production paths cover different parts of that work. The next test is a new-source artifact with a real correction, recorded costs and human judgment of the delivered result.

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